

Surname	Centre Number	Candidate Number
Other Names		0


GCSE – NEW

3445UB0-1


APPLIED SCIENCE (Double Award)
Unit 2: Space, Health and Life
HIGHER TIER

FRIDAY, 16 JUNE 2017 – MORNING

1 hour 30 minutes

	For Examiner’s use only		
	Question	Maximum Mark	Mark Awarded
Section A	1. (a)(b)(c)(d)	19	
	1. (e)	6	
Section B	2.	14	
	3.	10	
	4.	10	
	5.	10	
	6.	6	
	Total	75	

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ADDITIONAL MATERIALS

In addition to this examination paper, you will require a separate Resource Folder, calculator, pencil and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 1(e) is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A periodic table is printed on page 16.

Section A

Answer **all** questions in the spaces provided.

Use the information in the separate Resource Folder to answer the following questions.

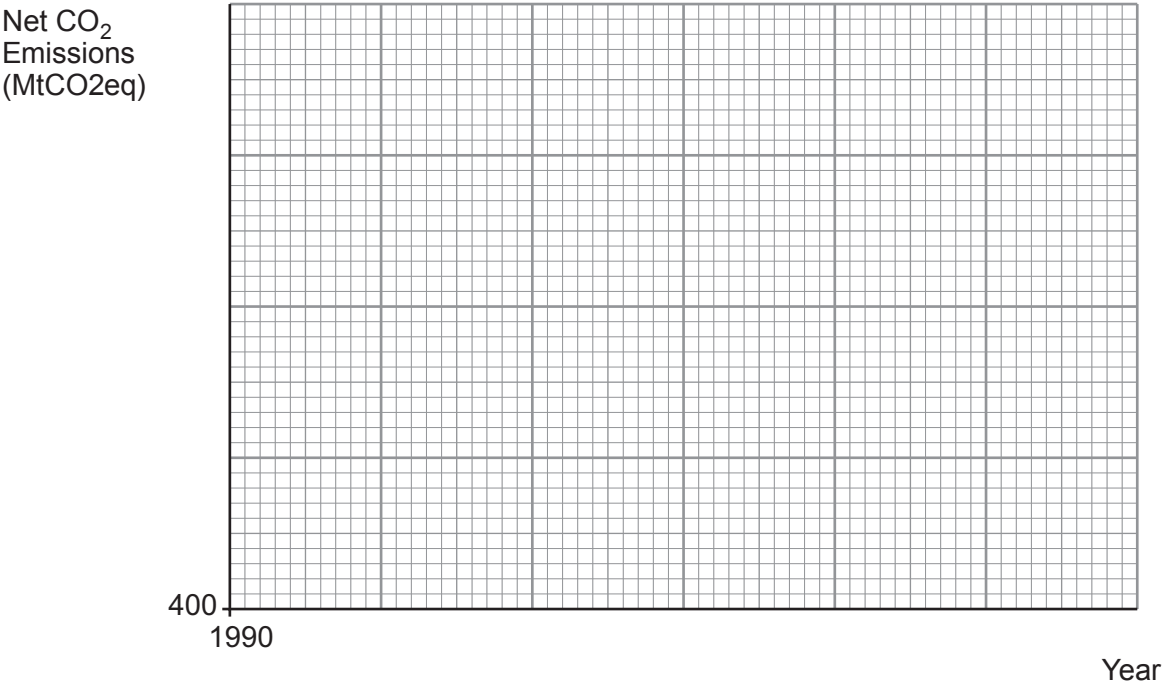
1. (a) (i) Use the information in **Table 1** to describe the trend in emissions of HFC between 1990 and 2014. [2]

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- (ii) Use the information in **Table 1** to plot a graph of net CO₂ emissions from the UK between 1990 and 2014. [4]



- (iii) Describe the trend shown in the graph. [2]

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- (iv) Describe how your graph (opposite) compares with the information in **Graph 1**. [2]

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- (v) Use your knowledge of the carbon cycle to account for the seasonal fluctuations in CO₂ concentration shown in **Graph 1**. Give reasons for your answer. [3]

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- (b) Use the information in **Tables 2** and **3** and the equation:

$$\text{greenhouse contribution} = (\text{concentration in the atmosphere}) \times (\text{multiplier})$$

to complete the table below.

[3]

gas	greenhouse contribution					
	multiplier	pre-industrial baseline	natural additions	man-made additions	total relative contribution	% of total contribution
carbon dioxide	1	288 000		11 880	368 400	72.37
methane	21	17 808	12 117		36 645	

- (c) Use the information in **Tables 2** and **4** to describe the effect on the data in the last column of **Table 2** when water vapour is included. [2]

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- (d) State **one** conclusion that can be made from the information shown in **Chart 1** on page 4 of the resource folder. [1]

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(e) Explain whether **Graph 2** and **Graph 3** support the view that global warming is due to solar activity and not CO₂ concentration. [6 QER]

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Section B

Answer all the questions in the spaces provided.

2. A class of students were using dice to model radioactive decay. There were 10 groups of students in the class. Each group of students had 90 dice.



- (a) (i) Describe how they should carry out the experiment.

[3]

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- (ii) State the number of dice you would expect to land with a 6 facing up on the first throw in **each** group.

[1]

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- (iii) Each group's results were added together to give the class results. Give **one** advantage of using a bigger sample size.

[1]

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(b) The table shows the results for the **whole class**.

Throw	Number of sixes	Number of dice remaining
0	0	900
1	146	754
2	124	630
3	103	527
4		442
5	72	370
6	61	309
7	52	257
8	38	219
9	34	

(i) **Complete** the table.

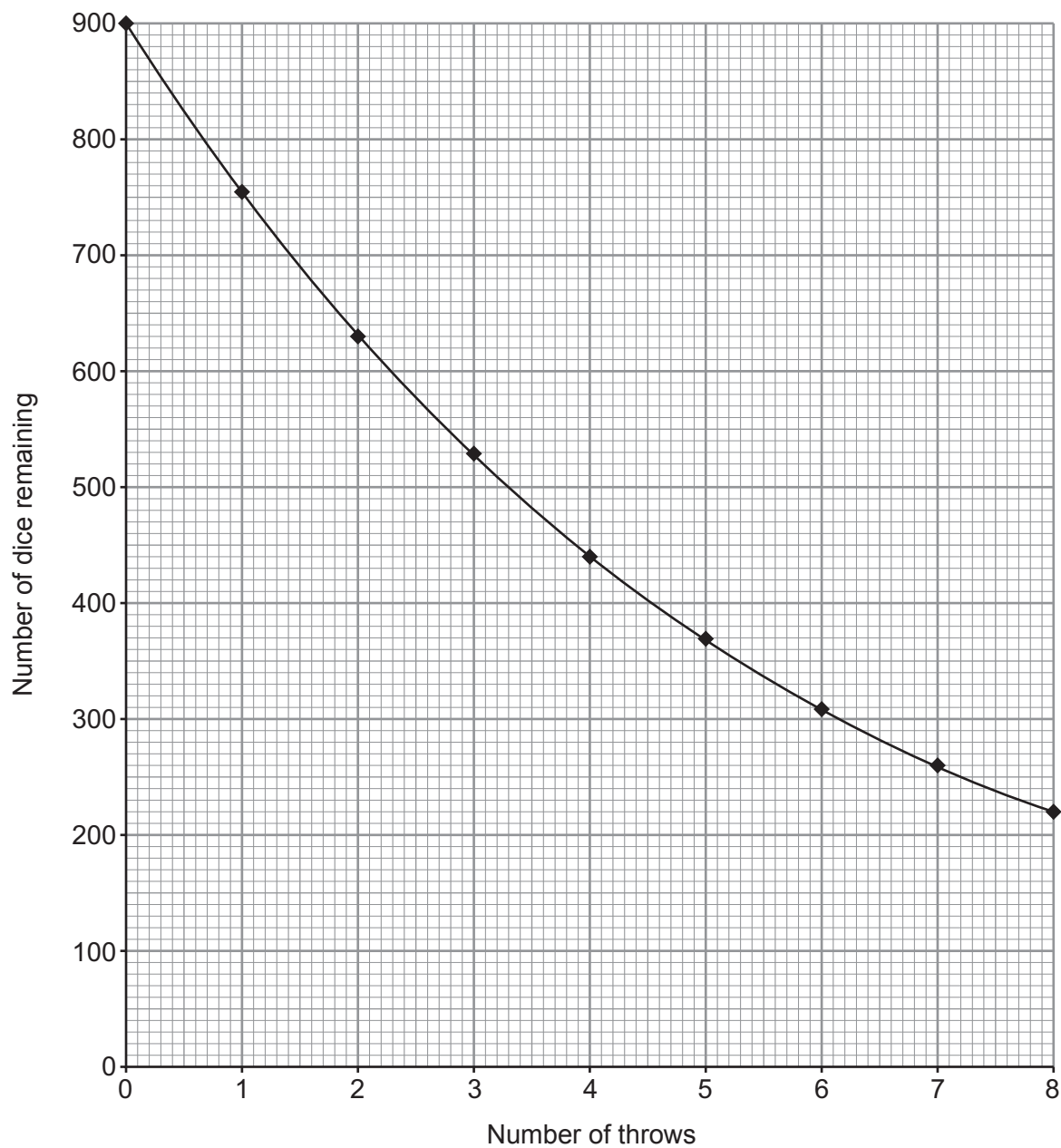
[2]

(ii) Use the data in the table to estimate the half-life of the 'dice'.

[1]

half-life = throws

(c) Some of the data in the table is plotted on the grid below.



(i) **Draw lines** on your graph to find the half-life of the dice.

[2]

half-life = throws

(ii) State **one** reason why finding the half-life using the graph is a better method than the one used in part (b)(ii).

[1]

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- (d) Radioisotopes have many medical uses. One use is to sterilise blood products for transfusion.

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- (i) Select from the table below the most suitable radioisotope for this purpose.

[1]

Radioisotope	Half-life	Type of emissions
bismuth-213	46 minutes	alpha
chromium-51	28 days	gamma
caesium-137	30 years	beta and gamma
bismuth-209	2×10^{19} years	alpha

- (ii) Give **two** reasons for your choice.

[2]

1.
2.

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3. Huntington's disease (HD) is an inherited disease that causes nerve cells in the brain to degenerate. People are born with the defective gene, but symptoms usually don't appear until middle age. Early symptoms of HD may include uncontrolled movements, clumsiness or balance problems. The disease is caused by an inherited defect in a single gene. With only one copy of the defective dominant allele (**H**), a person can develop the disease.

(a) State the meaning of the following terms:

[3]

(i) allele

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(ii) dominant allele

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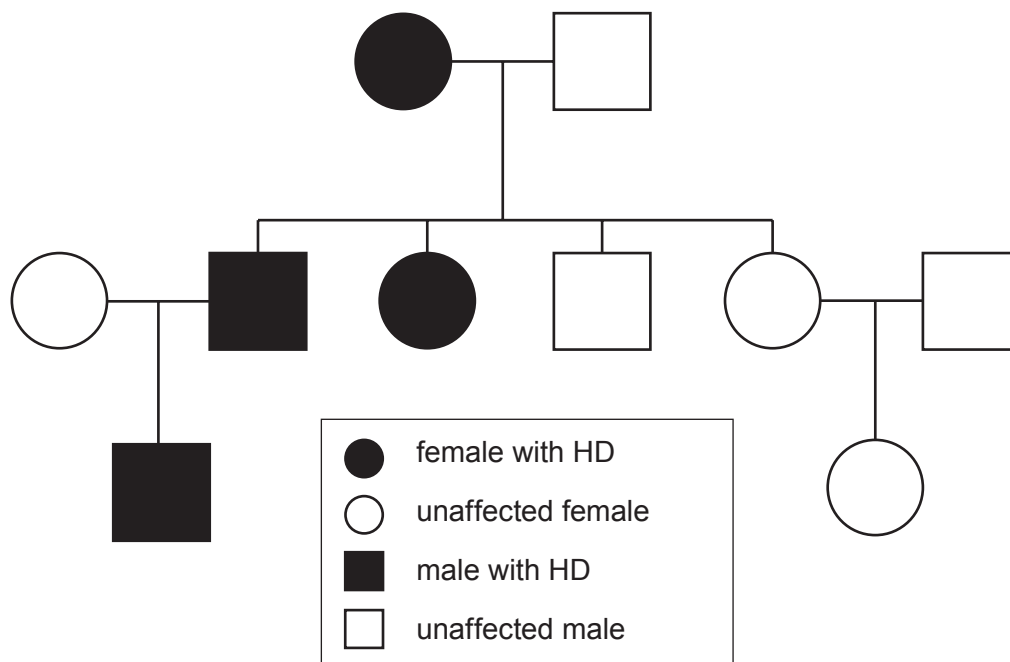
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(iii) genotype

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- (b) The diagram shows a family tree identifying those who are unaffected (do not have HD) and those with HD.



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Draw **two** Punnett squares and **compare** the percentage chances of a child developing HD if:

- one parent is unaffected and one parent is heterozygous
- both parents are heterozygous

[5]

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(c) Discuss the ethical problems which may arise when an individual undergoes genetic screening and finds out they have the allele for HD. [2]

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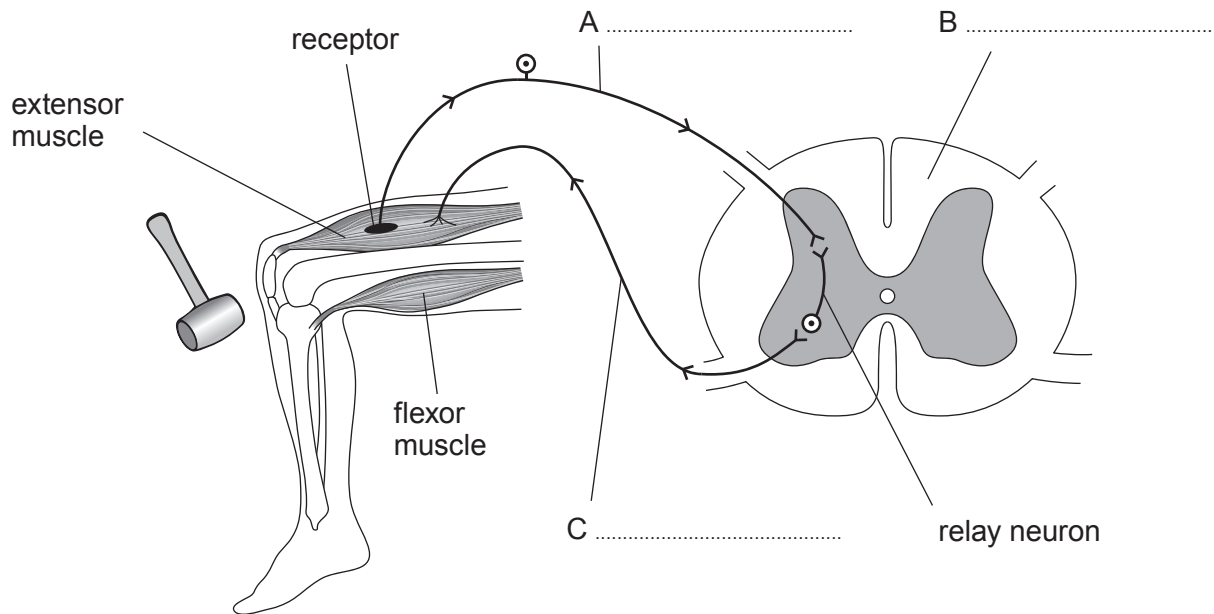
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4. The diagram shows a reflex arc.

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(i) **Label** the diagram. [3]

(ii) A doctor checks the reflexes of a patient by tapping the knee sharply which should result in the lower leg moving up quickly. Describe how the tap on the knee results in the lower leg moving up. Refer to the information in the diagram in your answer. [4]

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(iii) Describe the function of the antagonistic pair of muscles shown in the diagram. [3]

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5. Since October 2011 there has been a minimum charge of 5p for carrier bags in Wales.

(a) (i) Explain in terms of our natural resources why we should reduce our use of plastic carrier bags. [2]

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(ii) Describe **three** benefits to the environment if fewer plastic bags are used. [3]

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(b) Before the introduction of charges, 350 million plastic bags were used every year. Five years later this figure had dropped to 70 million used every year. Before a charge was introduced in England, 7 500 million plastic carrier bags were used per year. If the same success is achieved as in Wales, calculate the number of plastic carrier bags that will be in use per year in England five years later. [3]

number =

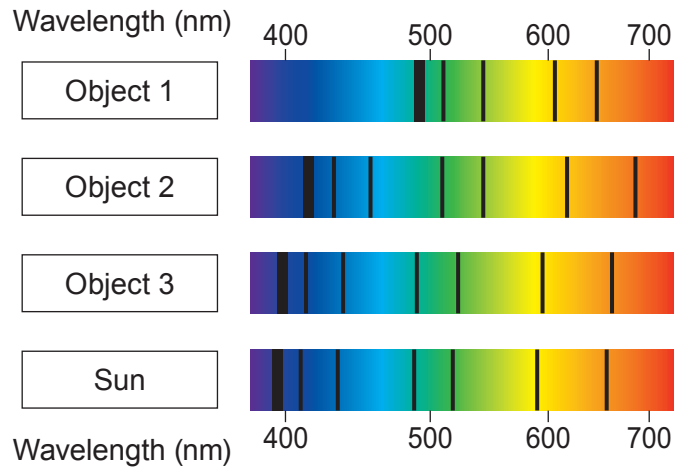
(c) When researchers collected reused plastic bags from supermarket shoppers they found large numbers of bacteria in almost all carrier bags. Explain why this is a concern. [2]

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6. The diagram below shows absorption spectra from four different objects in space. The bottom spectrum is from our Sun.



- (a) Explain how an absorption spectrum is produced.

[3]

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- (b) Explain how you can deduce, from the spectra, that objects 1, 2 and 3 have the same chemical composition as the Sun and are moving away from Earth at different speeds.

[3]

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1												4 He Helium 2						
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10	
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18	
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36	
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54	
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86	
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																

Key

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number